C4 Notes: trigonometric functions

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In stage B, sine, cosine and tangent were ratios of sides in a right triangle, so they only made sense for angles between $0°$ and $90°$. Here they become **functions defined for every real number**, which makes them the natural language for anything that repeats: tides, sound, the seasons, a swinging pendulum, alternating current. Dutch books call this subject *goniometrie*.

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1. Radians

A full turn is $360°$, a number chosen by the Babylonians. Mathematics prefers a unit that comes from the circle itself.

> **Definition.** The size of an angle in **radians** (*radialen*) is the length of the arc it cuts from a circle of radius 1 centred at the vertex.

The circumference of the unit circle is 2π, so

$$360° = 2\pi \text{ rad}, \qquad 180° = \pi \text{ rad}, \qquad 1 \text{ rad} = \frac{180°}{\pi} \approx 57.3°.$$

To convert degrees to radians multiply by $\frac{\pi}{180}$; back, multiply by $\frac{180}{\pi}$.

degrees	30	45	60	90	120	135	150	180	270	360
radians	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π

Arc length and sector area. In a circle of radius r, an angle of θ radians cuts off an arc of length $r\theta$ (scale the unit-circle arc by r) and a sector of area $\frac{1}{2}r^2\theta$ (the fraction $\frac{\theta}{2\pi}$ of the area πr^2). These clean formulas are the first reason to use radians.

> **Going further: the real reason for radians.** For small x in radians, $\sin x \approx x$: the vertical half-chord and the arc are almost equal. In degrees you would get $\sin x° \approx \frac{\pi}{180}x$, an ugly factor that would then appear in every derivative. In stage D you will prove $(\sin x)' = \cos x$, which is true **only** in radians.

From now on, angles are in radians unless a $°$ is written. Set your calculator to RAD.

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2. The unit circle

Let P be the point on the unit circle (*eenheidscirkel*) $x^2 + y^2 = 1$ reached by turning from $(1, 0)$ through an angle t, **anticlockwise** for $t > 0$ and clockwise for $t < 0$. Then define

$$\cos t = x_P, \qquad \sin t = y_P, \qquad \tan t = \frac{\sin t}{\cos t} \ \ (\cos t \neq 0).$$

For $0 < t < \frac{\pi}{2}$ this agrees with the triangle definitions: drop a perpendicular from P to the x-axis; the hypotenuse is 1, so the opposite side is $\sin t$ and the adjacent side is $\cos t$.

Immediate consequences:

- $-1 \leq \sin t \leq 1$ and $-1 \leq \cos t \leq 1$.
- **Periodicity:** turning a further 2π gives the same point, so $\sin(t + 2\pi) = \sin t$ and $\cos(t + 2\pi) = \cos t$.

- **Pythagoras:** P lies on $x^2+y^2=1$, so $\sin^2 t+\cos^2 t=1$ for every t. (Notation: $\sin^2 t$ means $(\sin t)^2$.)
- $\tan t$ is the slope of the line OP. Since P and $-P$ (turning a further π) give the same line, $\tan(t+\pi)=\tan t$: the period of tan is π.

Signs per quadrant. sin is positive in quadrants I and II (upper half), cos in I and IV (right half), tan in I and III.

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3. Exact values and symmetry

The two special triangles from stage B give the values in the first quadrant:

t	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin t$	0	$\frac{1}{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{3}$	1
$\cos t$	1	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}$	0
$\tan t$	0	$\frac{1}{3}\sqrt{3}$	1	$\sqrt{3}$	undefined

(Memory aid: sin goes $\frac{1}{2}\sqrt{0}, \frac{1}{2}\sqrt{1}, \frac{1}{2}\sqrt{2}, \frac{1}{2}\sqrt{3}, \frac{1}{2}\sqrt{4}$.)

Dutch exams write $\frac{1}{2}\sqrt{2}$ rather than $\frac{1}{\sqrt{2}}$; use that form.

Symmetries of the unit circle give all other values:

Symmetry	Formula
reflection in the x-axis	$\sin(-t)=-\sin t$, $\cos(-t)=\cos t$
reflection in the y-axis	$\sin(\pi-t)=\sin t$, $\cos(\pi-t)=-\cos t$
point reflection in O	$\sin(t+\pi)=-\sin t$, $\cos(t+\pi)=-\cos t$
reflection in $y=x$	$\sin\left(\frac{\pi}{2}-t\right)=\cos t$, $\cos\left(\frac{\pi}{2}-t\right)=\sin t$

So sin is odd and cos is even.

Example 1. $\cos\frac{5\pi}{6}$. The angle $\frac{5\pi}{6}=\pi-\frac{\pi}{6}$ is in quadrant II, where cos <0. So $\cos\frac{5\pi}{6}=-\cos\frac{\pi}{6}=-\frac{1}{2}\sqrt{3}$.

Example 2. $\sin\left(-\frac{11\pi}{4}\right)$. Add 4π: $-\frac{11\pi}{4}+4\pi=\frac{5\pi}{4}=\pi+\frac{\pi}{4}$, in quadrant III. $\sin\frac{5\pi}{4}=-\sin\frac{\pi}{4}=-\frac{1}{2}\sqrt{2}$.

Method: (1) reduce to $[0,2\pi)$ by adding multiples of 2π; (2) find the reference angle in the first quadrant; (3) fix the sign by the quadrant.

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4. Graphs

Unrolling the unit circle along the t-axis gives the graphs.

- $y=\sin x$: period 2π, zeros at $k\pi$, maximum 1 at $\frac{\pi}{2}+k2\pi$, minimum -1 at $\frac{3\pi}{2}+k2\pi$. Odd.
- $y=\cos x$: the sine graph shifted $\frac{\pi}{2}$ to the **left**, because $\cos x=\sin\left(x+\frac{\pi}{2}\right)$. Even.
- $y=\tan x$: period π, zeros at $k\pi$, vertical asymptotes at $x=\frac{\pi}{2}+k\pi$, increasing on each branch.

Sinusoids

Using the transformations from C1, every graph of the form

$$y = a + b\sin\big(c(x-d)\big) \qquad (b>0,\ c>0)$$

is a **sinusoid** (*sinusoïde*) with

- **equilibrium** (*evenwichtsstand*) $y=a$: the middle line;
- **amplitude** b: the distance from the middle line to a maximum;
- **period** $\dfrac{2\pi}{c}$: because $c(x-d)$ must grow by 2π for one full cycle;
- **starting point** (*beginpunt*) (d,a): a point on the middle line where the graph goes **up**.

So the maximum is $a+b$ and the minimum is $a-b$. For $y=a+b\cos(c(x-d))$, the point $(d,a+b)$ is a **maximum**.

Example 3. $f(x)=2+3\sin\big(2\,(x-\tfrac{\pi}{4})\big)$: equilibrium 2, amplitude 3, period π, starting point $(\tfrac{\pi}{4},2)$. Maximum 5 at $x=\tfrac{\pi}{4}+\tfrac{1}{4}\pi=\tfrac{\pi}{2}$ (a quarter period after the start), minimum -1 at $x=\pi$.

> **Common mistake.** Reading the horizontal shift from $\sin(2x-\tfrac{\pi}{2})$ as $\tfrac{\pi}{2}$. Factor out c first: $2x-\tfrac{\pi}{2}=2\,(x-\tfrac{\pi}{4})$, so the shift is $\tfrac{\pi}{4}$.

A formula from a graph

Example 4. A sinusoid has a maximum at $(2,5)$ and the next minimum at $(6,-1)$.

- Equilibrium $a=\frac{5+(-1)}{2}=2$; amplitude $b=\frac{5-(-1)}{2}=3$.
- Half a period is $6-2=4$, so the period is 8 and $c=\frac{2\pi}{8}=\frac{\pi}{4}$.
- A maximum at $x=2$ suggests a cosine: $y=2+3\cos\big(\frac{\pi}{4}(x-2)\big)$.
- Or with sine: the starting point (middle, going up) is a quarter period, 2, before the maximum, at $x=0$: $y=2+3\sin\big(\frac{\pi}{4}x\big)$.

Both formulas are correct; there are infinitely many (d can be shifted by any number of periods).

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5. Equations with all solutions

Because sine and cosine are periodic, an equation like $\sin x=\frac{1}{2}$ has **infinitely many** solutions. On the unit circle there are two points with $y=\frac{1}{2}$: at $t=\frac{\pi}{6}$ and at $t=\pi-\frac{\pi}{6}=\frac{5\pi}{6}$. Every solution is one of these plus a whole number of turns:

$$\sin x=\tfrac{1}{2} \iff x=\tfrac{\pi}{6}+k\cdot 2\pi \ \text{ or } \ x=\tfrac{5\pi}{6}+k\cdot 2\pi \qquad (k\in\mathbb{Z}).$$

The general rules (learn these; they are what the CE expects):

$$\sin A=\sin B \iff A=B+k\cdot 2\pi \ \text{ or } \ A=\pi-B+k\cdot 2\pi$$
$$\cos A=\cos B \iff A=B+k\cdot 2\pi \ \text{ or } \ A=-B+k\cdot 2\pi$$
$$\tan A=\tan B \iff A=B+k\cdot \pi$$

Why: two angles have the same sine iff their points on the unit circle have the same height, i.e. are equal or mirror images in the y-axis (that mirror sends B to $\pi-B$). For cosine, the mirror is the x-axis (sending B to $-B$). For tangent, the same line through O: the same point or the opposite point ($B+\pi$).

Method.

1. Bring the equation to the form $\sin A = \sin B$ (etc.), writing the number as a sine of a known angle: $\sin A = \frac{1}{2}\sqrt{3}$ becomes $\sin A = \sin \frac{\pi}{3}$.
2. Apply the rule and solve each case for x. **Divide the $k \cdot 2\pi$ term too.**
3. If an interval is given, list the solutions in it by trying $k = 0, \pm 1, \pm 2, \ldots$

Example 5. Solve $\cos 3x = \frac{1}{2}$ on $[0, \pi]$. $\cos 3x = \cos \frac{\pi}{3} \iff 3x = \frac{\pi}{3} + k2\pi$ or $3x = -\frac{\pi}{3} + k2\pi \iff x = \frac{\pi}{9} + k\frac{2\pi}{3}$ or $x = -\frac{\pi}{9} + k\frac{2\pi}{3}$. In $[0, \pi]$: from the first, $\frac{\pi}{9}, \frac{7\pi}{9}$; from the second, $\frac{5\pi}{9}$ (with $k = 1$). **Answer:** $x = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$. (Check the count: $\cos 3x$ has period $\frac{2\pi}{3}$, so on an interval of length π there are one and a half periods, and about three solutions. Good.)

Example 6. $\sin(2x) = \cos x$. Rewrite the right side: $\cos x = \sin\left(\frac{\pi}{2} - x\right)$. Then $2x = \frac{\pi}{2} - x + k2\pi \Rightarrow x = \frac{\pi}{6} + k\frac{2\pi}{3}$, or $2x = \pi - \left(\frac{\pi}{2} - x\right) + k2\pi \Rightarrow x = \frac{\pi}{2} + k2\pi$.

> **Common mistake.** Writing $2x = \frac{\pi}{3} + k2\pi$ and then $x = \frac{\pi}{6} + k2\pi$. You must divide everything by 2: $x = \frac{\pi}{6} + k\pi$. Otherwise you lose half the solutions.

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6. Identities

An **identity** (*formule, identiteit*) is an equation that holds for all values of the variable. You already know $\sin^2 t + \cos^2 t = 1$ and $\tan t = \frac{\sin t}{\cos t}$.

The difference formula for cosine

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta.$$

Proof. Take the points $A = (\cos\alpha, \sin\alpha)$ and $B = (\cos\beta, \sin\beta)$ on the unit circle. The angle between OA and OB is $\alpha - \beta$.

Compute AB^2 with coordinates:

$$AB^2 = (\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2 = 2 - 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta),$$

using $\cos^2 + \sin^2 = 1$ twice.

Compute AB^2 after rotating: rotate the whole picture by $-\beta$. Distances do not change. Now B goes to $(1, 0)$ and A to $(\cos(\alpha - \beta), \sin(\alpha - \beta))$. So

$$AB^2 = (\cos(\alpha - \beta) - 1)^2 + \sin^2(\alpha - \beta) = 2 - 2\cos(\alpha - \beta).$$

Comparing the two expressions gives the formula. □

Everything else follows

- Replace β by $-\beta$ (using $\cos(-\beta) = \cos\beta$, $\sin(-\beta) = -\sin\beta$):
$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta.$$
- Use $\sin\theta = \cos\left(\frac{\pi}{2} - \theta\right)$: $\sin(\alpha + \beta) = \cos\left(\left(\frac{\pi}{2} - \alpha\right) - \beta\right) = \cos\left(\frac{\pi}{2} - \alpha\right)\cos\beta + \sin\left(\frac{\pi}{2} - \alpha\right)\sin\beta$, so
$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta, \qquad \sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta.$$
- Put $\beta = \alpha$: the **double-angle formulas**
$$\sin 2\alpha = 2\sin\alpha\cos\alpha, \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 2\cos^2\alpha - 1 = 1 - 2\sin^2\alpha.$$

The CE formula sheet (*formulekaart*) lists these, but knowing where they come from means you can never misremember a sign.

Example 7. $\sin\frac{\pi}{12} = \sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \frac{1}{2}\sqrt{3}\cdot\frac{1}{2}\sqrt{2} - \frac{1}{2}\cdot\frac{1}{2}\sqrt{2} = \frac{\sqrt{6}-\sqrt{2}}{4}$.

Example 8 (equation needing an identity). Solve $\cos 2x = \cos x$ on $[0, 2\pi]$. Method 1 (identity): $2\cos^2 x - 1 = \cos x$, so $2\cos^2 x - \cos x - 1 = 0$, $(2\cos x + 1)(\cos x - 1) = 0$, $\cos x = -\frac{1}{2}$ or $\cos x = 1$. So $x = \frac{2\pi}{3}, \frac{4\pi}{3}$ or $x = 0, 2\pi$. Method 2 (rule): $2x = x + k2\pi$ or $2x = -x + k2\pi$, so $x = k2\pi$ or $x = k\frac{2\pi}{3}$. In $[0, 2\pi]$: $0, \frac{2\pi}{3}, \frac{4\pi}{3}, 2\pi$. Same answer.

Writing $a\sin x + b\cos x$ as one sinusoid. $\sin x + \cos x = \sqrt{2}\left(\frac{1}{2}\sqrt{2}\sin x + \frac{1}{2}\sqrt{2}\cos x\right) = \sqrt{2}\left(\sin x\cos\frac{\pi}{4} + \cos x\sin\frac{\pi}{4}\right) = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right)$. So the sum of a sine and a cosine with the same period is again a sinusoid, here with amplitude $\sqrt{2}$. In general $a\sin x + b\cos x = \sqrt{a^2 + b^2}\,\sin(x + \varphi)$ for a suitable φ.

Lowering the power. From the double-angle formulas: $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$ and $\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos 2x$. So $\sin^2 x$ is a sinusoid too, with period π.

> **Common mistake.** $\sin(\alpha + \beta) = \sin\alpha + \sin\beta$. Test: $\alpha = \beta = \frac{\pi}{2}$ gives $\sin\pi = 0$ on the left and 2 on the right.

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7. Sine rule and cosine rule

In any triangle ABC, label the sides opposite A, B, C as a, b, c and the angles α, β, γ.

Area formula

Drop the altitude from C to line AB. Its length is $h = b\sin\alpha$ (also when α is obtuse, because $\sin(\pi - \alpha) = \sin\alpha$). So

$$\text{area} = \tfrac{1}{2}\,c\,h = \tfrac{1}{2}\,bc\sin\alpha.$$

Sine rule

$$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma}.$$

Proof. The area is $\frac{1}{2}bc\sin\alpha = \frac{1}{2}ac\sin\beta = \frac{1}{2}ab\sin\gamma$ (same formula from each vertex). Divide all three by $\frac{1}{2}abc$: $\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{c}$. □

Problem 23 shows that the common value is $2R$, the diameter of the circumscribed circle.

Use: when you know two angles and a side, or two sides and an angle opposite one of them.

> **Common mistake: the ambiguous case.** Given a, b and α, the sine rule gives $\sin\beta$; but β and $\pi - \beta$ have the same sine. Both may give a valid triangle. Check whether $\alpha + (\pi - \beta) < \pi$.

Cosine rule

$$c^2 = a^2 + b^2 - 2ab\cos\gamma.$$

Proof. Place C at the origin and CB along the positive x-axis: $C = (0,0)$, $B = (a,0)$, $A = (b\cos\gamma, b\sin\gamma)$ (this is the unit circle definition, scaled by b, so it works for obtuse γ too). Then

$$c^2 = AB^2 = (b\cos\gamma - a)^2 + (b\sin\gamma)^2 = b^2(\cos^2\gamma + \sin^2\gamma) - 2ab\cos\gamma + a^2 = a^2 + b^2 - 2ab\cos\gamma. \quad \square$$

For $\gamma = \frac{\pi}{2}$ this is Pythagoras. For $\gamma > \frac{\pi}{2}$, $\cos\gamma < 0$ and $c^2 > a^2 + b^2$: an obtuse angle faces a longer side than Pythagoras would give.

Use: two sides and the angle between them (find the third side), or three sides (find an angle).

Example 9. A triangle has sides $a = 5$, $b = 7$, $c = 8$. Find the angle β opposite the side 7, and the area. By the cosine rule $b^2 = a^2 + c^2 - 2ac\cos\beta$, so $\cos\beta = \dfrac{5^2 + 8^2 - 7^2}{2 \cdot 5 \cdot 8} = \dfrac{40}{80} = \dfrac{1}{2}$, so $\beta = \frac{\pi}{3}$ (60°). Area $= \frac{1}{2} \cdot 5 \cdot 8 \cdot \sin\frac{\pi}{3} = 10\sqrt{3}$.

The ambiguous case in numbers

Example 10. In triangle ABC: $a = 7$, $b = 10$, $\alpha = 35°$. Find β. Sine rule: $\sin\beta = \dfrac{b\sin\alpha}{a} = \dfrac{10\sin 35°}{7} \approx 0.819$. The calculator gives $\beta \approx 55.0°$, but $\beta \approx 180° - 55.0° = 125.0°$ has the same sine.

- $\beta \approx 55.0°$: $\gamma \approx 90.0°$, a valid triangle.
- $\beta \approx 125.0°$: $\gamma \approx 20.0°$, also valid (since $35° + 125.0° < 180°$).

So there are **two** triangles with these data, with $c \approx 12.20$ and $c \approx 4.18$. Draw it: with A and side $b = AC$ fixed, a circle of radius 7 around C meets the other ray from A in two points. If a had been $\geq b$, only one triangle would exist.

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8. Inverse trigonometric functions and modelling

Inverse functions

On the whole real line sin is not one-to-one, so it has no inverse. Restricted to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ it is increasing and takes every value in $[-1,1]$ once; the inverse is arcsin (on the calculator $\sin^{-1}$). Similarly arccos : $[-1,1] \to [0,\pi]$ and arctan : $\mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$.

The calculator gives only **one** solution of $\sin x = 0.3$, namely $\arcsin 0.3 \approx 0.305$. The other family comes from the rule: $x = \pi - 0.305 + k2\pi \approx 2.837 + k2\pi$. Never stop at the calculator answer.

Modelling a periodic process

Example 11. In the Netherlands, the number of hours of daylight on day t of the year is roughly modelled by

$$D(t) = 12.3 + 4.5\sin\left(\frac{2\pi}{365}(t - 80)\right).$$

(The model is a simplification; the numbers are illustrative.) Equilibrium 12.3 h around 21 March ($t = 80$), maximum 16.8 h a quarter year later, minimum 7.8 h.

On how many days is there more than 15 hours of daylight? $D(t) > 15 \iff \sin u > 0.6$ with $u = \frac{2\pi}{365}(t - 80)$. In one period: $\arcsin 0.6 < u < \pi - \arcsin 0.6$, i.e. $0.6435 < u < 2.4981$. Converting back: $117.4 < t < 225.1$, an interval of about 107.7 days. So on about 108 days.

Going further: why $\sin x \approx x$ for small x. For $0 < x < \frac{\pi}{2}$, compare three areas in the unit circle: the triangle O, $(1,0)$, $(\cos x, \sin x)$ has area $\frac{1}{2}\sin x$; the sector has area $\frac{1}{2}x$; the right triangle O, $(1,0)$, $(1, \tan x)$ has area $\frac{1}{2}\tan x$. Each contains the previous one, so

$$\sin x < x < \tan x.$$

Dividing by $\sin x$ and inverting: $\cos x < \dfrac{\sin x}{x} < 1$. As $x \to 0$, $\cos x \to 1$, so $\dfrac{\sin x}{x} \to 1$. This limit is the key step in proving $(\sin x)' = \cos x$ in stage D, and the sector area $\frac{1}{2}x$ is exactly where radians are needed. Numerically: $\sin 0.1 \approx 0.09983$, $\tan 0.1 \approx 0.10033$.

Common mistakes, collected

- Calculator in DEG mode when the problem is in radians (or the reverse).
- Giving only the calculator solution of $\sin x = c$ and forgetting $\pi - x$.
- Dividing by $\sin x$ or $\cos x$ in an equation, which loses the solutions where it is 0. Factor instead: $\sin x \cos x = \sin x \iff \sin x(\cos x - 1) = 0$.
- Forgetting to divide $k \cdot 2\pi$ by the coefficient of x.
- $\sin 2x = 2\sin x$ (wrong) instead of $\sin 2x = 2\sin x \cos x$.
- Ignoring the second triangle in the ambiguous case of the sine rule.

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9. A complete CE-style example

Example 12. A Ferris wheel has diameter 40 m; its axle is 22 m above the ground. It turns anticlockwise once every 8 minutes. At $t = 0$ a passenger is at the lowest point.

a. Give a formula for the height $h(t)$. Equilibrium 22, amplitude 20, period 8, so $c = \frac{2\pi}{8} = \frac{\pi}{4}$. At $t = 0$ the passenger is at a minimum: $h(t) = 22 - 20\cos\left(\frac{\pi}{4}t\right)$.

b. Bereken exact during how many minutes per turn the passenger is higher than 32 m. $22 - 20\cos\left(\frac{\pi}{4}t\right) = 32 \iff \cos\left(\frac{\pi}{4}t\right) = -\frac{1}{2} \iff \frac{\pi}{4}t = \frac{2\pi}{3} + k2\pi$ or $\frac{\pi}{4}t = -\frac{2\pi}{3} + k2\pi$. In the first turn ($0 \le t \le 8$): $t = \frac{8}{3}$ and $t = -\frac{8}{3} + 8 = \frac{16}{3}$. Between these the passenger is above 32 m (around the maximum at $t = 4$). Duration $\frac{16}{3} - \frac{8}{3} = \frac{8}{3}$ minutes, i.e. 2 minutes 40 seconds.

Summary

- Radians: arc length on the unit circle; π rad $= 180°$; arc $r\theta$, sector $\frac{1}{2}r^2\theta$.
- $\cos t$, $\sin t$ are the coordinates of the point at angle t on the unit circle.
- Exact values from the special triangles plus symmetry plus the sign per quadrant.
- $y = a + b\sin(c(x - d))$: equilibrium a, amplitude b, period $\frac{2\pi}{c}$, starting point (d, a).
- $\sin A = \sin B$: $A = B + k2\pi$ or $A = \pi - B + k2\pi$; $\cos A = \cos B$: $A = \pm B + k2\pi$; tan: $+k\pi$.
- Everything follows from $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$.
- Sine rule from the area formula; cosine rule from coordinates.